

Dynamic Macroeconomics II

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Log-linearization and Uhlig's method

Second Part

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Undetermined Coefficients

What's next:

- We have a linear stochastic system of equations.
- Now we need to solve it.
- Solving it means finding the **policy functions**.
- I.e. write endogenous variables as functions of state variables, of which there are exogenous and endogenous.
- The method is called Undetermined Coefficients because we do not know the values of parameters linking states to endogenous variables, so we need to find them.
- Uhlig proposes to gather equations into two groups:
 - 1 "Static" equations that do not involve expectations.
 - 2 Dynamic equations that include expectations.
- I first present his general notation.
- Then we write the Hansen example we have been working with into Uhlig's notation: "fill matrices".
- To understand what Uhlig's **codes** do, we do Undetermined Coefficients by hand.

Undetermined Coefficients: general notation

- Let x_t be the $m \times 1$ **endogenous state variables** vector, y_t the $n \times 1$ **other endogenous variables** vector and z_t the $k \times 1$ **exogenous variables** vector (bad notation, we used z for a single productivity shock before).
- The relation between x_t , y_t and z_t can be written as:

$$0 = Ax_t + Bx_{t-1} + Cy_t + Dz_t$$

$$0 = E_t [Fx_{t+1} + Gx_t + Hx_{t-1} + Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t]$$

$$z_{t+1} = Nz_t + \varepsilon_{t+1}$$

$$0 = E_t [\varepsilon_{t+1}]$$

- We need to find matrices P , Q , R y S in **policy functions**:

$$x_t = Px_{t-1} + Qz_t$$

$$y_t = Rx_{t-1} + Sz_t$$

- Uhlig has codes that solve for P , Q , R , S as functions of $A, B, C, D, F, G, H, J, K, L, M, N$.

Example: Uhlig's version of Hansen's model

- Main question: which are the endogenous, and the exogenous, state variables?

$$\begin{aligned} y_t &= \begin{bmatrix} c_t \\ y_t \\ n_t \\ \lambda_t \\ r_t \end{bmatrix}_{5 \times 1} \\ z_t &= \begin{bmatrix} z_t \end{bmatrix}_{1 \times 1} \\ x_t &= \begin{bmatrix} k_t \end{bmatrix}_{1 \times 1}. \end{aligned}$$

- **Notice Uhlig's notation:** x_t is the endogenous state variable chosen at time t : k_t in his version of Hansen's.
- Uhlig decided to include Lagrange multiplier λ_t in the system, could have eliminated this unknown.
- Next, obtain matrices $A, B, C, D, F, G, H, J, K, L, M, N$.

- Set log-linearized system equal to zero:

$$0 = c_t + \lambda_t$$

$$0 = -n_t + y_t + \lambda_t$$

$$0 = -\bar{R}r_t + \rho \frac{\bar{Y}}{\bar{K}} (y_t - k_{t-1})$$

$$0 = -y_t + z_t + \rho k_{t-1} + (1 - \rho) n_t$$

$$0 = -\bar{C}c_t - \bar{K}k_t + \bar{Y}y_t + (1 - \delta) \bar{K}k_{t-1}$$

$$0 = E_t [-\lambda_t + \lambda_{t+1} + r_{t+1}]$$

$$0 = -z_{t+1} + \psi z_t + \varepsilon_{t+1}.$$

- Fill the matrices.

$$A = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -\bar{K} \end{bmatrix}_{5 \times 1}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ -\rho \frac{\bar{Y}}{\bar{K}} \\ +\rho \\ + (1-\delta) \bar{K} \end{bmatrix}_{5 \times 1}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & \rho \frac{\bar{Y}}{\bar{K}} & 0 & 0 & -\bar{R} \\ 0 & -1 & (1-\rho) & 0 & 0 \\ -\bar{C} & \bar{Y} & 0 & 0 & 0 \end{bmatrix}_{5 \times 5}$$

$$D = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}_{5 \times 1}$$

$$F = \begin{bmatrix} 0 \end{bmatrix}_{1 \times 1}$$

$$G = \begin{bmatrix} 0 \end{bmatrix}_{1 \times 1}$$

$$H = \begin{bmatrix} 0 \end{bmatrix}_{1 \times 1}$$

$$J = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \end{bmatrix}_{1 \times 5}$$

$$K = \begin{bmatrix} 0 & 0 & 0 & -1 & 0 \end{bmatrix}_{1 \times 5}$$

$$L = \begin{bmatrix} 0 \end{bmatrix}_{1 \times 5}$$

$$\begin{aligned} M &= \begin{bmatrix} 0 \end{bmatrix}_{1 \times 1} \\ N &= \begin{bmatrix} \psi \end{bmatrix}_{1 \times 1} \end{aligned}$$

- At this point we can use Uhlig's codes to solve for P, Q, R, S
- But first let's apply it by hand to see what it does.

- Recall log-linearized equations:

$$-c_t = \lambda_t \quad (1)$$

$$n_t = y_t + \lambda_t \quad (2)$$

$$\bar{R}r_t = \rho \frac{\bar{Y}}{\bar{K}} (y_t - k_{t-1}) \quad (3)$$

$$y_t = z_t + \rho k_{t-1} + (1 - \rho) n_t \quad (4)$$

$$\bar{C}c_t + \bar{K}k_t = \bar{Y}y_t + (1 - \delta) \bar{K}k_{t-1} \quad (5)$$

$$\lambda_t = E_t[\lambda_{t+1} + r_{t+1}] \quad (6)$$

$$z_t = \psi z_{t-1} + \varepsilon_t. \quad (7)$$

Uhlig writes the system as a function of only two variables: the multiplier and capital

For example

- Take n_t from (2), plug into (4), solve for y_t to get:

$$y_t = \frac{1}{\rho} z_t + k_{t-1} + \frac{1-\rho}{\rho} \lambda_t \quad (8)$$

Write (5) and (6) eliminating c_t , y_t , r_t

U_{b0} y_t y divide both ends K

- Use the following notation:

$$\alpha_1 = \frac{\bar{Y}}{\bar{K}} + (1 - \delta)$$

$$\alpha_2 = \frac{\bar{C}}{\bar{K}} + \frac{1 - \rho}{\rho} \frac{\bar{Y}}{\bar{K}}$$

$$\alpha_3 = \frac{\bar{Y}}{\rho \bar{K}}$$

$$\alpha_4 = 0$$

$$\alpha_5 = 1 + (1 - \rho) \frac{\bar{Y}}{R\bar{K}}$$

$$\alpha_6 = \frac{\bar{Y}}{R\bar{K}}.$$

- Rewrite feasibility condition and Euler equation as (**check**):

$$0 = -k_t + \alpha_1 k_{t-1} + \alpha_2 \lambda_t + \alpha_3 z_t \quad (9)$$

$$0 = E_t [-\lambda_t + \alpha_4 k_t + \alpha_5 \lambda_{t+1} + \alpha_6 z_{t+1}]. \quad (10)$$

$$0 = -k_t + \alpha_1 k_{t-1} + \alpha_2 \lambda_t + \alpha_3 z_t \quad y_t = \frac{y}{k} \left(\frac{1}{\rho} z_t + k_{t-1} + 1 - \frac{1}{\rho} \right) d_t$$

$$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$$

$$\frac{y}{k} + (1-\delta) \quad \quad \quad \frac{c}{k} + 1 - \frac{1}{\rho} \frac{y}{k}$$

$$-k_t + \left[\frac{y}{k} + (1-\delta) \right] k_{t-1} + \left(\frac{c}{k} + 1 - \frac{1}{\rho} \frac{y}{k} \right) \lambda_t + \frac{y}{\rho k} z_t$$

$$\lambda_t = E_t [\lambda_{t+1}, r_{t+1}] \quad r_t = \rho \frac{y}{kR} (y_t - k_{t-1})$$

$$0 = E_t \left[-\lambda_t + \alpha_4 k_t + \alpha_5 \lambda_{t+1} + \alpha_6 z_{t+1} \right]$$

$$r_{t+1} = \rho \frac{y}{kR} \left(\frac{1}{\rho} z_{t+1} + k_t + 1 - \frac{1}{\rho} \lambda_{t+1} - k_t \right)$$

$$\downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow$$

$$0 \quad \quad \quad 1 + (1-\rho) \frac{y}{kR} \quad \quad \quad \frac{y}{kR}$$

$$-\lambda_t = E_t \left[\lambda_{t+1} + (1-\rho) \frac{y}{kR} \lambda_{t+1} + \frac{y}{kR} z_{t+1} \right]$$

- Propose a solution to the model.
- State variables are k_t and z_t , while control variable is λ_t .
Then:

$$k_t = \eta_{KK} k_{t-1} + \eta_{KZ} z_t$$

$$\lambda_t = \eta_{\lambda K} k_{t-1} + \eta_{\lambda Z} z_t$$

- The η s are unknown coefficients, need to find them.
- Substitute into (9) and (10), and use the fact that

$$E_t(z_{t+1}) = \psi z_t \text{ to get (check!): } 0 = -k_t + \alpha_1 k_{t-1} + \alpha_2 \lambda_t + \alpha_3 z_t$$

$$0 = E_t[-\lambda_t + \alpha_4 k_t + \alpha_5 \lambda_{t+1} + \alpha_6 z_{t+1}]$$

$$0 = (-\eta_{KK} + \alpha_1 + \alpha_2 \eta_{\lambda K}) k_{t-1} \\ + (-\eta_{KZ} + \alpha_2 \eta_{\lambda Z} + \alpha_3) z_t$$

$$0 = (-\eta_{\lambda K} + \alpha_4 \eta_{KK} + \alpha_5 \eta_{\lambda K} \eta_{KK}) k_{t-1} \\ + (-\eta_{\lambda Z} + \alpha_4 \eta_{KZ} + \alpha_5 \eta_{\lambda K} \eta_{KZ} + (\alpha_5 \eta_{\lambda Z} + \alpha_6) \psi) z_t$$

$$\downarrow$$

$$\psi z_t + \cancel{z_t}$$

- Both equations must hold simultaneously for all k_{t-1} , z_t .
- Therefore, coefficients must be zero.
- For k_{t-1} :

$$0 = -\eta_{KK} + \alpha_1 + \alpha_2\eta_{\lambda K}$$

$$0 = -\eta_{\lambda K} + \alpha_4\eta_{KK} + \alpha_5\eta_{\lambda K}\eta_{KK}.$$

- To compute η_{KK} , solve for $\eta_{\lambda K}$ from the first equation and substitute into the second to get a quadratic equation (**check**):

$$0 = \eta_{KK}^2 - \left(\alpha_1 - \frac{\alpha_2}{\alpha_5}\alpha_4 + \frac{1}{\alpha_5} \right) \eta_{KK} + \frac{\alpha_1}{\alpha_5}$$

- Then use solution to a quadratic equation:

$$\eta_{KK} = \frac{1}{2} \left(\left(\alpha_1 - \frac{\alpha_2}{\alpha_5} \alpha_4 + \frac{1}{\alpha_5} \right) \pm \sqrt{\left(\alpha_1 - \frac{\alpha_2}{\alpha_5} \alpha_4 + \frac{1}{\alpha_5} \right)^2 - 4 \frac{\alpha_1}{\alpha_5}} \right)$$

- We can show that (**check**)

$$\eta_{KK,1} \eta_{KK,2} = \frac{\alpha_1}{\alpha_5} = \bar{R} = \frac{1}{\beta} > 1$$

- Roots are almost reciprocal, except for the β :

$$\eta_{KK,1} = \frac{1}{\beta \eta_{KK,2}}$$

- At most one root is stable, i.e. < 1 .
- This is a characteristic of the neoclassical growth model. See Ljungqvist & Sargent (2004), or Stokey and Lucas, with Prescott (1989).
- Obtain $\eta_{\lambda K}$ with η_{KK} .

- Calculate the value of the coefficients for z_t :

$$0 = -\eta_{Kz} + \alpha_2 \eta_{\lambda z} + \alpha_3$$

$$0 = -\eta_{\lambda z} + \alpha_4 \eta_{Kz} + \alpha_5 \eta_{\lambda K} \eta_{Kz} + (\alpha_5 \eta_{\lambda z} + \alpha_6) \psi.$$

- Since the value of $\eta_{\lambda K}$ is already known, we have a linear system of two equations and two unknowns, which we can solve:

$$\begin{aligned} \eta_{\lambda z} &= \frac{\alpha_4 \alpha_3 + \alpha_5 \eta_{\lambda K} \alpha_3 + \alpha_6 \psi}{1 - \alpha_4 \alpha_2 - \alpha_5 \eta_{\lambda K} \alpha_2 - \alpha_5 \psi} \\ \eta_{Kz} &= \alpha_2 \eta_{\lambda z} + \alpha_3. \end{aligned}$$

- We have applied by hand Uhlig's method to find the (coefficients of the) policy functions.
- I.e. we have solved the model.
- Next: Show how to use Uhlig's code to solve models in general.